Creating heterogeneous enzyme-metal hybrid catalysts (EMHCs) using candida antarctica lipase B (CALB) enzymes and ruthenium nanoparticles to improve the synthesis of enantiopure chiral amines

Introduction

- **Enantiomers** are a pair of molecules that exist in two forms that are mirror images of each other with a left and right hand side.
- A **racemic mixture** has equal amounts of both enantiomers.
- Different enantiomer sides typically exhibit different biological activities, pharmacological properties, and toxicities. Therefore, enantiomerically pure amines are very important molecules in the production of **pharmaceuticals**, **agrochemicals**, and **special chemical industries**.
- Conventional routes to create them require **harsh conditions,** such as high temperatures and pressures, and compromise catalyst recovery.
- **Heterogeneous catalysts** are more reusable and create a complementary, enhanced effect by keeping the different components of the catalyst separate.

Image taken from Indo Amines Limited

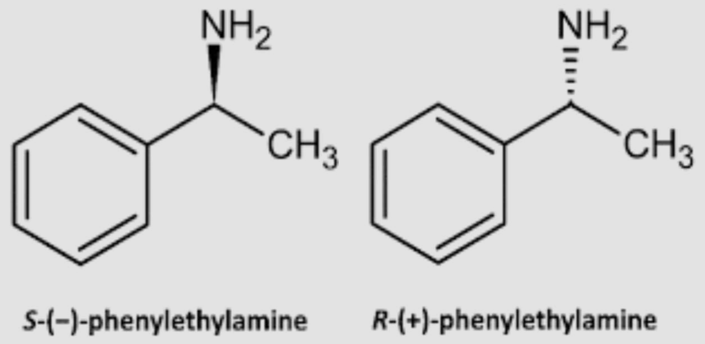


Image taken from Mamontova et al. (2020)

Overview & Purpose

- This project aims to combine both a **metal** (ruthenium) and an **enzyme** (candida antarctica lipase B) catalyst, immobilized on a **heterogenous base** (magnetic wrinkled organosilica) to **deracemize racemic phenylethylamine**.
- **Novel combinations** of enzymes and metals allow for a more comprehensive understanding of how enzymes and metals influence one another's catalytic performance, impacting stereoselectivity and activity.
- Increase opportunities for **future applications** of EMHCs as new combinations may lead to more variety in development options and improvements in catalyst recovery.

IV: The presence of the **CALB enzyme** in the heterogeneous EMHC.	**DV:** The **enantiopurity** of phenylethylamine measured by **enantiomeric excess percentage** and **conversion yield**	**Control:** Ru@MWOS followed with CALB for DKR (seperate steps)

Hypothesis: If an EMHC with ruthenium nanoparticles and CALB enzymes is tested on racemic phenylethylamine, then it will result in greater enantiomeric excess percentages compared to the ruthenium catalyst, signaling more enantiomerically pure products. This is because the EMHC will have enhanced, complementary features with benefits such as high activity and enantioselectivity of both enzymatic biocatalysis and metal-catalysis.

$$\frac{actual\ yield}{theoretical\ yield} \times 100\%$$

Conversion Yield

$$ee\ (\%) = \frac{(A_{R-iso} - A_{S-iso})}{(A_{R-iso} + A_{S-iso})} \times 100\%$$

Excess Percentage

Methodology

1. Make Fe_3O_4 Nanoparticles

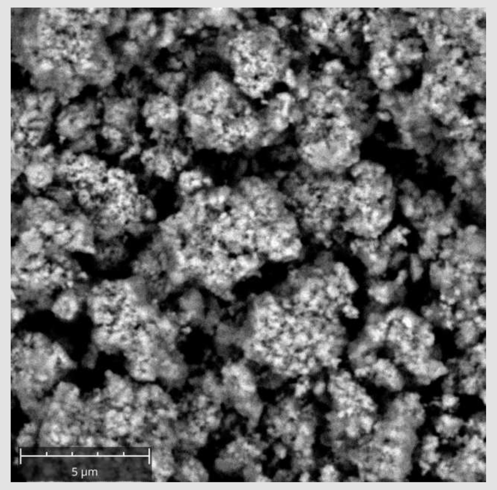


Image taken from Dong et al. (2023)

2. Prepare and synthesize the **co-immobilization carrier**, magnetic wrinkled organosilica, and immobilize **RuNP** and **CALB.**

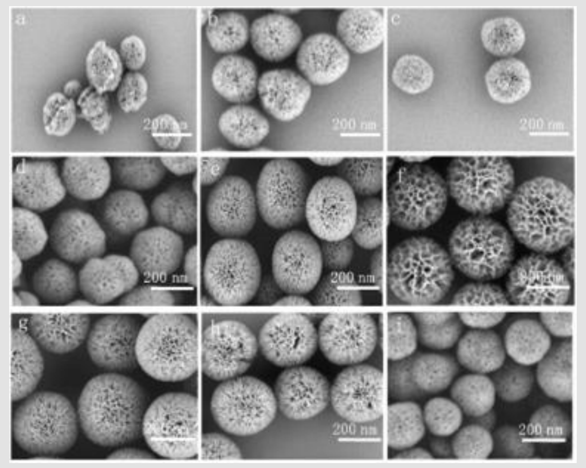


Image taken from Dong et al. (2023)

3. Test the EMHC on **phenylethylamine**

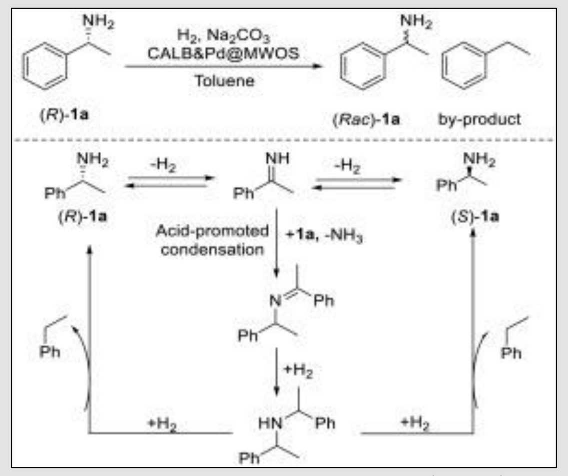


Image taken from Dong et al. (2023)

4. Analyze the catalytic activity and enantiopurity of the products using **NMR** and **Mosher's Method**

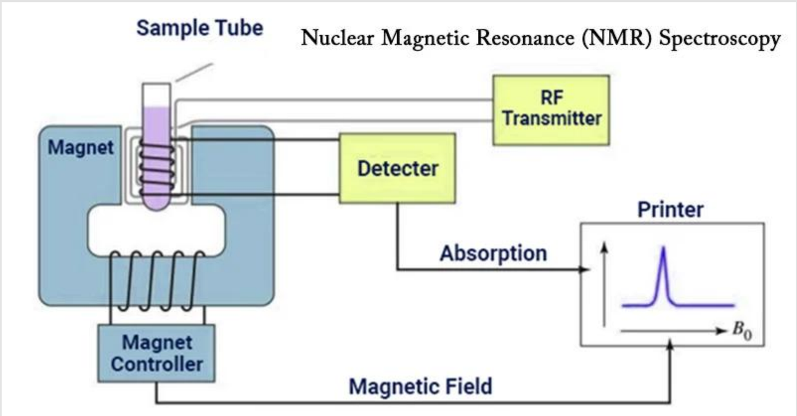


Image taken from Sagar (2022)

5. Separate and recover the heterogeneous catalysts using a **magnet** to test **reusability**

Image taken from BBC Bitesize (2023)

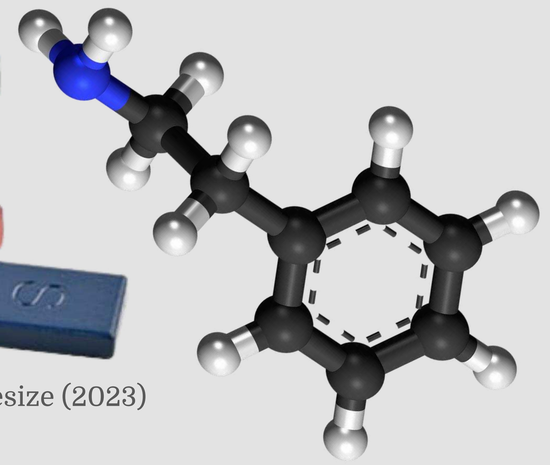
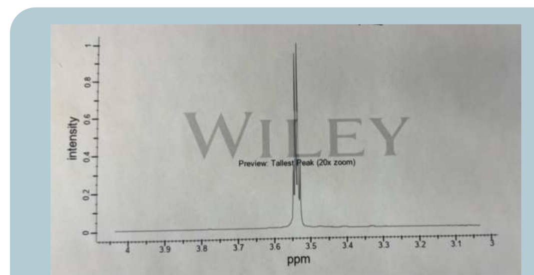
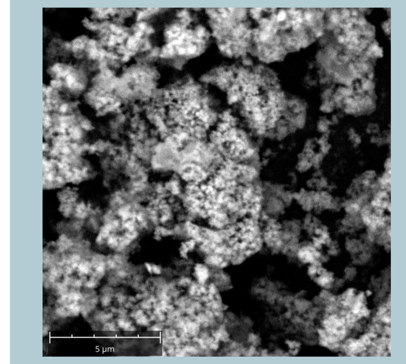


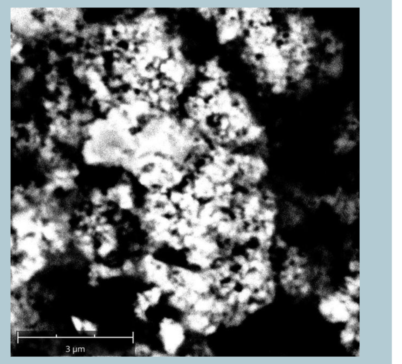
Image taken from Jynto (2011)

Results

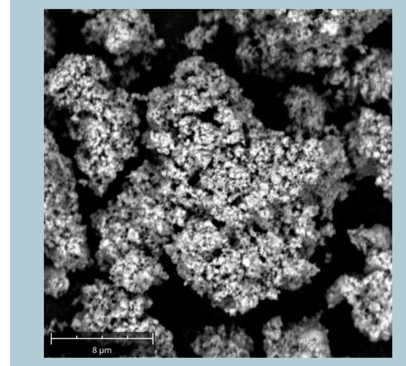


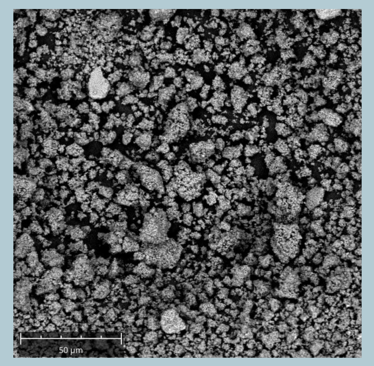
HNMR Characterization of (-)-alpha-Methoxy-alpha-(trifluoromethyl)phenylacetyl chloride



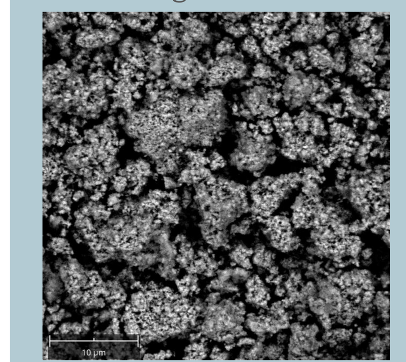


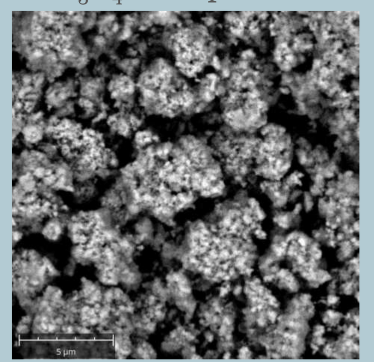
SEM Images of Fe_3O_4 Nanoparticles



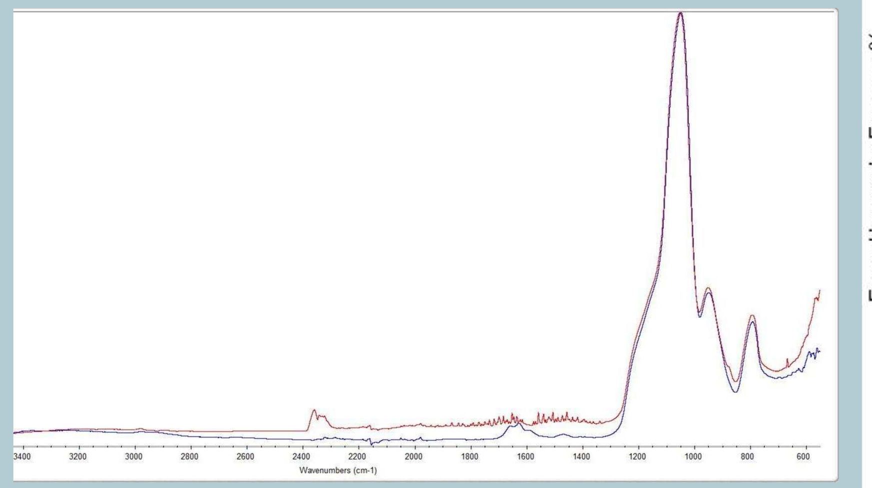


SEM Images of Silica Coated Fe_3O_4 Nanoparticles

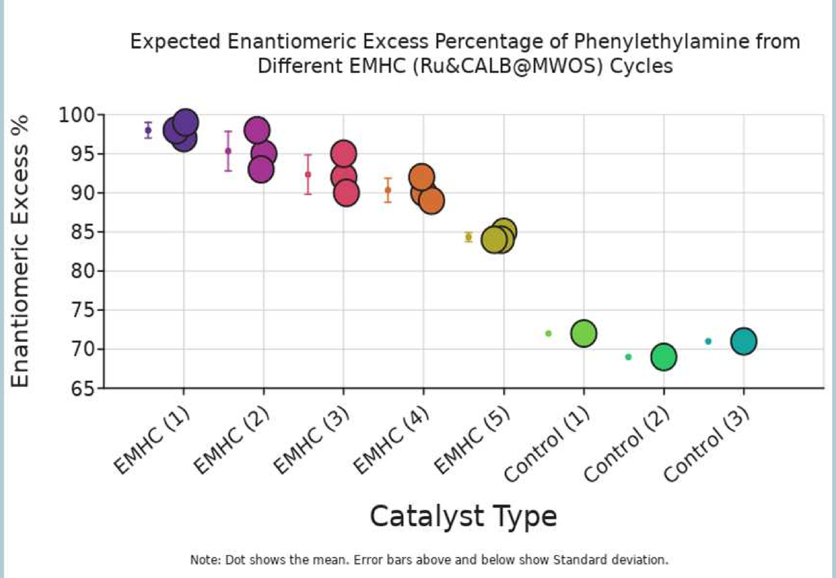




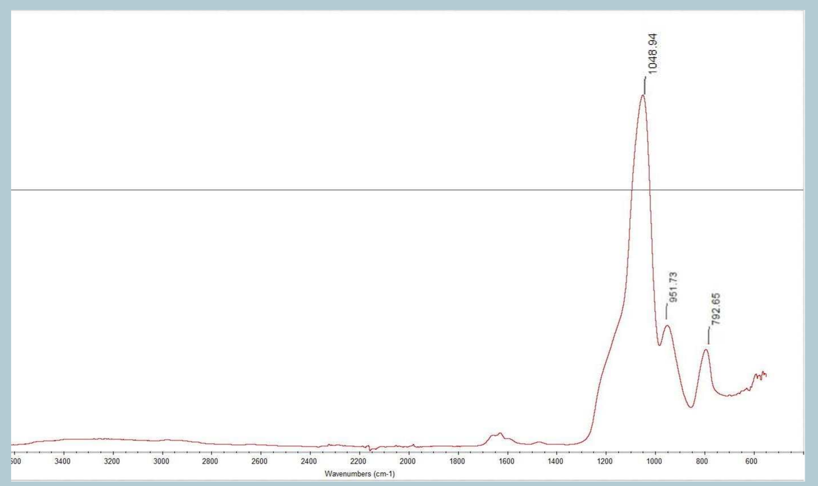
SEM Images of Synthesized MWOS Carrier



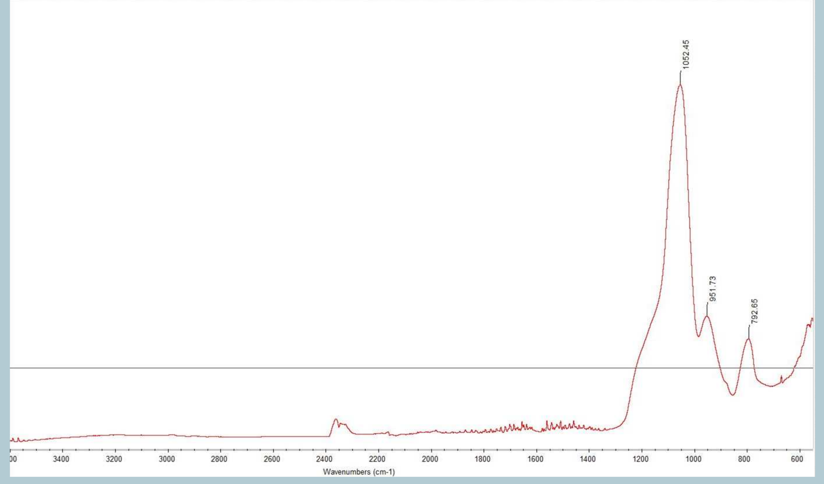
FTIR Comparison of Fe_3O_4 Nanoparticles and MWOS Carrier
Fe_3O_4 Nanoparticles: Red
MWOS Carrier: Blue



Expected Data



FTIR Characterization of MWOS Carrier
- $1048.94\ cm^{-1}$ peak show Si-O-Si stretch confirming the successful formation of the silica framework in the MWOS support
- $\sim570\ cm^{-1}$ peak around 600 confirms the presence of FeO bonds



FTIR Characterization of Fe_3O_4 Nanoparticles
- $1052.45\ cm^{-1}$ peak show Si-O-Si stretch confirming the successful formation of the silica framework in the MWOS support
- $\sim580\ cm^{-1}$ peak around 600 confirms the presence of FeO bonds

Conclusion/Future Work

- **FTIR** and **SEM** characterizations of the iron oxide core has shown the presence of **FeO** bonds as well as the presence of **silica**. The FTIR peaks correspond with the presence of Si-O and Fe-O bonds.
- When the silica coating was added to the Fe_3O_4 nanoparticles, the SEM images showed the presence of a **lighter exterior** compared to the **dark, iron interior**.
- The SEM characterization of the carrier has shown how it has grown by adding another **layer**, resulting in clearer images. However, as we are working with nanoparticles, it is difficult to obtain clear images.
- This year, the **synthesis** of the heterogeneous **MWOS** (Magnetic Wrinkled Organosilica) carrier was completed as well as the immobilization of the ruthenium nanoparticles onto the carrier. As this was planned to be a 2 year project, the focus for next year will be to **collect data** from testing the finished catalyst on phenylethylamine.
- In the **experimental trials**, the catalyst is set to interact with the enantiopure chiral amine, phenylethylamine, for a set amount of time, making the following steps quicker compared to making the catalyst.
- **Reusability** of the enzyme and EMHC will also be tested by conducting cycles of testing on the same catalyst.
- The amount of trials are limited by the amount of catalysts made and amount of material available.

References

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